Classtest-2. Digital Logic design. Time : 30 min. Marks: 20
Answer any 2(two) of the following questions:

1.	Define the basic comparator operation. Design a two bit comparator (truth table, min. Boolean expression and circuit diagram).	10
2.	How Decoders work? Draw the circuit diagram with waveforms for a 3:6 decoder. Show how decoder is used to identify a I/O device in a computer system?	10
3.	a) Implement the following Boolean expression using 4:1 and 8:1 multiplexer block separately. $\overline{A}BC + \overline{A}\overline{B}C + A\overline{B}C + ABC + AB\overline{C}$	5
	b) Implement a 16:1 multiplexer using 4:1 and 8:1 Multiplexer blocks separately.	5